UNIVERSITY OF THE PUNJAB

Third Semester 2015
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Elementary Mathematics-II (Calculus) **TIME ALLOWED: 2 hrs. & 30 mins.**
Course Code: MATH-211 **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

	Short Questions	
Q2	I. Evaluate $\int (3x^6 - 2x^2 + 7x + 1)dx$.	10×2
	II. Differentiate $\dfrac{x^2+2x+1}{x^2-2x+1}$ with respect to x.	
	III. Evaluate $\int_0^1 \dfrac{dx}{x^2+9}$.	
	IV. Find the second derivative $\dfrac{d^2y}{dx^2}$ if $y = 9x^4 + 6x^2 + 1$.	
	V. Evaluate $\lim\limits_{x\to 3} \dfrac{x-3}{x^2-9}$.	
	VI. Find the average rate of change of the function $f(x) = x^2$ over the interval $[-1, 1]$.	
	VII. Evaluate $\int \ln x\, dx$	
	VIII. For what value of k $\lim\limits_{x\to 1} f(x)$ exists, where $f(x) = \begin{cases} 2kx, & \text{if } x < 1 \\ 6-2kx, & \text{if } x > 1 \end{cases}$	
	IX. Evaluate $\int_0^{2\pi} \sin^2 x\, dx$	
	X. Evaluate $\int \dfrac{t\sqrt{t}+\sqrt{t}}{t^2}\, dt$	
	Long Questions	
Q3	If $y = \sqrt[3]{\dfrac{x(x+1)(x-2)}{(x^2+1)(2x+3)}}$, then find $\dfrac{dy}{dx}$ by using logarithmic differentiation.	10
Q4	a) Evaluate $\int e^x \sin x\, dx$ b) Evaluate $\lim\limits_{y\to 0} \dfrac{\sin 3y \cot 5y}{y \cot 4y}$.	10
Q5	a) $\int_{2/\sqrt{3}}^{2} \dfrac{\cos(\sec^{-1} x)dx}{x\sqrt{x^2-1}}$ b) $\int_0^{\pi/4} \sqrt{1+\cos 4x}\, dx$	10

UNIVERSITY OF THE PUNJAB

Third Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Elementary Mathematics-II (Calculus) **TIME ALLOWED: 2 hrs. & 30 mins.**
Course Code: MATH-211/MTH-21107 **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

Short Questions

Q2 **(5x4)**

I. Check the continuity of $f(x)=|x-3|$ at $x=3$.

II. Find dy/dx if $\sqrt{x}+\sqrt{y}=\sqrt{a}$

III. Evaluate $\int \tan x / \cos x+\sec x \, dx$.

IV. Evaluate $\int_1^3 \ln x \, dx$

V. Evaluate $\int_0^1 \tan^{-1} x \, dx$.

Long Questions

Q3 **(10)**

Let $f(x)= x \sin(1/x)$ if $x\neq0$ and $f(0)=0$. Discuss the continuity and differentiability of **f** at $x=0$.

Q4 **(10)**

Differentiation w.r.t X

$$Y= X^x e^x \sin x \, (\ln x)$$

Q5 **(10)**

(i)Find $d(e^{\sqrt{x}})/dx$ and hence evaluate $\int_0^1 e^{\sqrt{x}}/\sqrt{x} \, dx$.

(ii)Evaluate $\int \tan^2 x \, \mathrm{Sec}^4 x \, dx$.

UNIVERSITY OF THE PUNJAB

Third Semester 2018
Examination: B.S. 4 Years Programme

Roll No. ....

PER: Elementary Mathematics-II (Calculus) **TIME ALLOWED: 2 hrs.**
urse Code: MATH-211/MTH-21107 **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

Short Questions

Q2:write the answers of the following questions (5 X4)

i) Solve $x^2-3x>10$

ii) Evaluate $\lim_{x\to 0}\frac{\ln(\tan x)}{\ln x}$

iii) Find dy/dx if $y=x^{\ln x}$

iv) Evaluate $\int x\sec^2 x\,dx$

v) Evaluate $\int_0^1 \frac{dx}{(1-x)^{\frac{1}{3}}}$

Long Questions

Q3 **(10)**

Let $f(x)=|x|$ if $x\neq 0$ and $f(0)=0$.Discuss the continuity and differentiability of f at x=0

Q4

Differentiate w.r.t x (5,5)

a) $y=\sqrt{(x^3+\csc x)}$

b) $y=(1+x^5\cot x)^{-8}$

Q5

Evaluate

a) $\int \frac{x^2}{\sqrt{x^3+1}}dx$

b) $\int_0^{\pi/8}\sin^5 2x\cos 2x\;dx$ (5,5)

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No. ....

PAPER: Elementary Mathematics-II (Calculus)
Course Code: MATH-211/MTH-21107 Part – II

MAX. TIME: 2 Hrs. 30 Min.
MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2: write the answers of the following questions (5 x 4 = 20 Marks)

i) Define continuity and Solve $4+7x \leq 2x-10$

ii) Evaluate $\lim_{x\to 0} \frac{\ln(\tan x)^2}{\ln x}$

iii)Find dy/dx if $y=a^x$

iv)Evaluate $\int xe^x dx$

v) Evaluate $\int_2^1 \frac{dx}{(2-x)^{\frac{1}{3}}}$

Long Questions **(3x10=30)**

Q3

a)Solve $\frac{2x-5}{x-2} < 1$

b)Discuss the continuity of function $f(x)=x\sin(1/x)$ at $x=0$

Q4

Differentiate w.r.t x

a) $\sin(\sqrt{(1+\cos x)}\,)$

b) $y= x\ln x \cos^2 \pi x$

Q5

Evaluate

a) $\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx$

b) $\int_0^2 \frac{x}{\sqrt{1+x^3}} dx$

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program :Third Semester – 2020

Roll No. ....

Paper: Elementary Mathematics-II (Calculus) **Time: 2 Hrs. 30 Min. Marks: 50**

Course Code: MATH-211/MTH-21107 **Part – II**

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Solve the following: **(5x4=20)**

i. Solve $|x| + |x - 1| > 1$

ii. Evaluate $\lim_{n \to 0} \frac{sinax}{sinbx}$

iii. Find $\frac{dy}{dx}$ if $x^3 + y^3 - 3axy = 0$

iv. Evaluate $\int x \, See^2 x \, dx$

v. Evaluate $\int_1^2 \frac{x^2+1}{x+1}$

Solve the following: **(3x10=30)**

Q.3.

i. Solve $\frac{x^2 - 2}{1 - 2x} > 1$ (05)

ii. Find $\lim_{x \to 0} \frac{a^x - 1}{x}$ (05)

Q.4. Differentiate w.r.t 'x'

i. $y = e^{ax} Cos\ (barctanx)$ (05)

ii. $y = x \quad a^x Sinhx$ (05)

Q.5. Evaluate

i. $\int \frac{1}{\sqrt{a^2 - x^2}} \, dx$ (05)

ii. $\int_0^{\frac{\pi}{6}} x \, Cos\, x \, dx$ (05)

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Third Semester – Fall 2021

Roll No. ....

Paper: Elementary Mathematics-II (Calculus)
Course Code: MATH-211

Time: 3 Hrs. Marks: 60

Q.1. Solve the following: **(6x5=30)**

1. Find the domain of the real valued function given by $f(x) = \frac{1}{\sqrt{(1-x)(2-x)}}$.

2. Solve the equation $|x| + |x-1| = 0$ for $x < 0$.

3. Find the derivative of the function $5^{(\sin 3x)}$ with respect to x.

4. Evaluate the integral $\int x^2 e^x dx$.

5. Evaluate the definite integral $\int_{-1}^{5} |x-2| dx$.

6. Let the radius of the circle be increasing at the rate of 2 m/s. How fast is the area of that circle increasing when the radius of the circle is 40 m?

Q.2. Solve the following: **(3x10=30)**

1. Examine whether the given function is continuous at $x = 0$,

$$f(x) = \begin{cases} (1+3x)^{\frac{1}{x}}, & x \neq 0; \\ e^2, & x = 0. \end{cases}$$

2. Find $\frac{dy}{dx}$ by implicit differentiation for the the curve $\tan^{-1}\left(\frac{y}{x}\right) + yx^2 = 1$.

3. Evaluate $\int \frac{1}{3\sin x + 4\cos x} dx$.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Spring 2022

Roll No. ....

Paper: Elementary Mathematics-II (Calculus)
Course Code: MATH-211

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following: **(6x5=30)**

1. Find the domain of the real valued function given by $f(x) = \sqrt{4 - x^2}$.
2. Solve the equation $|2x| + |x - 3| = 0$ for $x < 0$.
3. Find the derivative of the function $e^{(x\cos 5x)}$ with respect to x.
4. Evaluate the integral $\int x^2 \ln x dx$.
5. Evaluate the definite integral $\int_0^{\pi} \sin 3x \cos 5x dx$.
6. Let the radius of the circle be increasing at the rate of 2 m/s. How fast is the area of that circle increasing when the radius of the circle is 40 m?

Q.2. Solve the following: **(3x10=30)**

1. Examine whether the given function is continuous at $x = 0$,

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0; \\ 2, & x = 0. \end{cases}$$

2. Find $\frac{dy}{dx}$ by implicit differentiation for the the curve $y\sin^{-1}(x) - x\tan^{-1}(y) = 1$.
3. Show that the function $f(x) = |x|$ is continuous but not differentiable at $x = 0$.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Fall 2022

Roll No. ....

Paper: Calculus-II Course Code: CAL-211 Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following: (6x5=30)

1. Find the polar form of the complex number $z = -2 + 2i$. Also find z^5.

2. Evaluate $\int 5x^2(x^5 + 12)dx$.

3. Evaluate $\int \frac{2x}{(8-x)^3}dx$ using integration by parts.

4. The marginal cost is given by $MC = 25 + 30Q - 9Q^2$. Fixed cost is given by $FC = 55$. Find the Total Cost, Average Cost and Variable Cost functions.

5. Given the demand function $P_d = 113 - Q^2$ and the supply function $P_s = (Q+1)^2$. Under pure competition, find the consumer surplus (CS) and producer's surplus (PS).

6. Solve the differential equation $\frac{dy}{dt} - 6y = 18$

Q.2. Solve the following: (3x10=30)

1. Derive the differential equation of Solow Growth Model for cost and labour problem in terms of single variable $\frac{K}{L}$.

2. Solve the following second order linear differential equation $y''(t) - 4y'(t) - 5y(t) = 35$ subject to the conditions $y(0) = 5$, $y'(0) = 6$.

3. Evaluate the integral $\int x^3 \sin x dx$.

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B.S. 4 Years Program : Third Semester – Spring 2023

Roll No.

Time: 3 Hrs. Marks: 60

Paper: Calculus-II Course Code: CAL-211

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(6x5=30)

Q.1. Solve the following:

1. Find the polar form of the complex number $z = 1 + 1i$. Also find z^5.
2. Find the particular solution, complementary function and general solution of the following difference equation $y_t + 7y_{t-1} + 6y_{t-2} = 42$.
3. Evaluate $\int \frac{3x}{(x^2-6)^5}\,dx$ using integration by parts.
4. The marginal cost is given by $MC = 25 + 30Q - 9Q^2$. Fixed cost is given by $FC = 55$. Find the Total Cost, Average Cost and Variable Cost functions.
5. Given the demand function $P_d = 25 - Q^3$ and the supply function $P_s = 2Q + 1$. Under pure competition, find the consumer surplus (CS) and producer's surplus (PS).
6. Solve the differential equation $\frac{dy}{dt} - 6y = 18$

Q.2. Solve the following: (3x10=30)

1. Solve the following second order linear differential equation $y''(t) + y'(t) + \frac{1}{4}y(t) = 9$ subject to the conditions $y(0) = 30$, $y'(0) = 15$.
2. Derive the differential equation of Solow Growth Model for cost and labour problem in terms of single variable $\frac{K}{L}$.
3. Evaluate the integral $\int x^3 e^x dx$.